

Analysing Water Potability to Support SDG 6 Clean Water and Sanitation & SDG 3 Good Health and Wellbeing

HW 2 Group 1

1. Model Selection & Justification

Random Forest was selected as the primary model because it handles non-linear feature interactions through ensemble decision trees, making it robust to the complex relationships between pH, sulfate, and potability. Logistic Regression served as a linear baseline for comparison.

The goal is to correctly identify unsafe water sources so that public health interventions can be prioritised in water stressed regions

2. Data Preparation

- The dataset contains 3,276 samples with 9 physicochemical features. Total missing values: 1,434. Rows with missing values were dropped, reducing the dataset to 2,011 rows. The original class distribution was imbalanced (1,200 not potable vs 811 potable), so the minority class was upsampled to 1,200 using sklearn's resample. All features were standardised with StandardScaler. For the merge,
- To contextualize the data within a global framework, we merged the potability data with the World Bank Water Stress dataset. The data measures how much of a country's available freshwater is consumed. However, the original potability dataset contains no geographic information. There are no country names so this made a geographically accurate merge impossible. The only available approach was to randomly assign country codes using np.random.seed(42). This was not a preferred choice but the merge was still performed to practise combining datasets in a real pipeline.

3. Model Performance Results

Model	Pre-merge Accuracy	Post-merge accuracy	AUC
Random first	87.50%	81.67%	~0.89
K-Nearest Neighbors	72.92%	70.00%	~0.78
Logistic Regression	50.83%	51.67%	~0.52
Decision Tree	73.75	74.79%	~0.75

4. Interpretation & SDG Insights

The results prove that water safety is a complex, non-linear problem. Random forest is the superior choice for identifying safe drinking sources.

The near-zero linear correlations explain why Logistic Regression fails. Water safety is determined by multi-feature interactions, not single parameters. Random Forest's ROC AUC of 0.89 versus Logistic Regression's 0.52 confirms this. Post-merge accuracy dropped for Random Forest (87.08% → 81.67%) because the randomly assigned water stress values introduced noise. Decision Tree marginally improved (+1%) likely due to overfitting to that noise.

The pre-upsampling run (unbalanced data) showed Logistic Regression predicting zero potable samples (recall of 0.00) confirming that class balancing is essential, not optional. Recall for Not Potable (0.85–0.90) being higher than for Potable (0.77) is acceptable in a public health context. Flagging unsafe water as safe is far more dangerous than the reverse.

For SDG 6 and SDG 3, countries with extreme water stress (UAE: 1,587%, Afghanistan: 54.76%) cannot afford to consume unsafe water. AI-driven classification from basic chemical readings enables rapid low cost screening, prioritising intervention where it is most needed.

5. Conclusion

- The random country code assignment means Water_Stress_2020 provides no real predictive signal. A better geographic dataset would be required to test whether regional water stress improves classification.
- Random oversampling duplicates existing samples. SMOTE would generate synthetic ones for more diverse training data.

Random Forest is the recommended model. It is the only classifier that can detect actual safe water. For SDG aligned deployment, models must identify safe water, not just predict the majority class. Advanced methods are a requirement, not a performance preference.